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ME 5130: Advanced Heat and Mass Transfer
Homework 03: External Flow
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5–9, Problem Statement: Consider application of the conservation principles of Chapter 2 to control volume $A-B-C-D$ for the steady laminar boundary-layer problem sketched in Fig. 5P-9.

PUT FIGURE IN

(a) Show that the mass flow rate across $B-C$ equals

$$\int_0^y (\rho U - \rho u)\, dy = \int_{x_A}^{x_D} \rho v(y)\, dy \tag{1}$$

(b) Show that the drag force exerted on the fluid by the plate

$$\int_0^y \frac{\rho U U}{g_c}\, dy - \left[\int_0^y \frac{\rho u u}{g_c}\, dy + \int_{x_A}^{x_B} \frac{\rho v(y) U}{g_c}\, dx\right] - F_{\text{drag plate on fluid}} = 0 \tag{2}$$

(c) Use the results of parts a and b to show that, when $y \to \infty$ to avoid shear stresses that exist near the plate,

$$F_{\text{drag}} = \frac{\rho}{g_c} \int_0^\infty u(U-u)\, dy \tag{3}$$

And

$$\delta_2 = x\frac{\bar{C}_f}{2} = 0.66412 \left(\frac{\nu x}{U}\right)^{\frac{1}{2}} \tag{4}$$

5–9, Solution:

5–13, Problem Statement:
Consider steady parallel flow of a constant-property fluid over a flat plate. When Pr/to0, the thermal boundary layer is much thicker than the velocity boundary layer, so that throughout the thermal boundary layer $F \approx \eta$ since then $u/U = F' \approx 1$. Such is the case for liquid metals.

(a) Show that in such a case Eq. (5–26) gives

$$\frac{d\theta(0)}{d\eta} = \left(\frac{\text{Pr}}{\pi}\right)^{\frac{1}{2}} \tag{5}$$

(b) Show that the result of part a gives, with $\text{Nu} = hx/k$ and $\text{Re} = Ux/\nu$,

$$\lim_{\text{Pr}\to 0} \frac{\text{Nu}}{\text{RePr}^{\frac{1}{2}}} = \frac{1}{0.33206\pi^{\frac{1}{2}}} \frac{C_f}{2} = 1.7\frac{C_f}{2} \tag{6}$$

5–13, Solution: